

A3-PS020-TC035

Local SRv6 Path Steering and Measured Failure Recovery

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Problem Description

The project aims to evaluate Local SRv6 Path Steering and Measured Failure Recovery in a controlled five-router IPv6/SRv6 virtual testbed.

- Configure two explicit SRv6 paths using ordered Segment Identifiers (SIDs).
- Introduce a controlled link failure and switch to a scripted backup policy. Verify SID traversal using packet captures.
- Compare explicit SRv6 path steering with ordinary hop-by-hop IPv6 routing on the same topology.
- Measure packet loss, interruption time, failure detection delay, repair delay, and SRv6 header overhead.
- Repeat the failure scenario to obtain reliable measurements.
- Investigate TI-LFA/BFD as a stretch goal.

The objective is to study and measure routing behaviour and recovery performance under controlled failure conditions without assuming a guaranteed operational advantage.

Background

Traditional IPv6 routing forwards packets hop-by-hop based on routing information maintained by the routers. The path is determined by routing protocols, which select the next hop according to the current network state.
Segment Routing over IPv6 (SRv6) enables an ingress router to steer traffic along an explicit path by encoding an ordered list of Segment Identifiers (SIDs) in the IPv6 packet's Segment Routing Header (SRH).
This project uses a controlled five-router IPv6/SRv6 virtual testbed to compare explicit SRv6 path steering with ordinary IPv6 routing. Two paths are configured: a primary path through R1 → R2 → R3 → R5 and a backup path through R1 → R2 → R4 → R5.
A controlled link failure is introduced to evaluate the behaviour of a scripted backup-policy switch. Packet captures are used to verify SID traversal, while measurements of packet loss, interruption time, failure detection delay, repair delay, and SRv6 header overhead help assess the recovery process.

Literature & RFC Review

Review of existing proposals, studies, and standards relevant to the project

1. RFC 8754 — IPv6 Segment Routing Header (SRH)

RFC 8754 defines the IPv6 Segment Routing Header used to carry segment-routing information in IPv6 packets. It provides the standards-based foundation for encoding an ordered list of Segment Identifiers (SIDs), enabling explicit path steering in the proposed S testbed.

2. AIORI-3 PS-020 Specification

The PS-020 problem specification defines the requirements for the five-router IPv6/SRv6 testbed, including two explicit paths, a controlled link failure, a scripted backup-policy swit packet-capture verification, and performance measurements.

Relevance to the Project

These references guide the implementation and evaluation of SRv6 path steering and failu recovery. Packet captures will verify SID traversal, while measurements of packet loss, interruption time, failure detection delay, repair delay, and SRv6 header overhead will be u to compare the SRv6 experiment against ordinary IPv6 routing on the same topology.

Proposed Solution

The proposed solution implements Local SRv6 Path Steering and Measured Failure Recovery using a five-router IPv6/SRv6 virtual testbed.

Two explicit SRv6 paths are configured using ordered Segment Identifiers (SIDs):

Primary Path: R1 → R2 → R3 → R5, used for normal traffic forwarding.

Backup Path: R1 → R2 → R4 → R5, activated after a controlled link failure.

Normal State: Traffic enters through R1 and follows the configured primary SRv6 policy.

Failure State: A selected network link is deliberately brought down to simulate a failure.

Recovery State: A scripted mechanism switches the ingress router's policy to the backup path to restore traffic forwarding.

Validation: Packet captures are analysed to confirm the expected SID traversal before and after the policy switch.

The experiment compares SRv6 path steering with ordinary hop-by-hop IPv6 routing using the same topology. Performance is evaluated through packet loss, interruption time, failure detection delay, repair delay, and SRv6 header overhead. Repeated trials are conducted to obtain reliable measurements.

Optimization Proposed

List of optimizations and enhancements

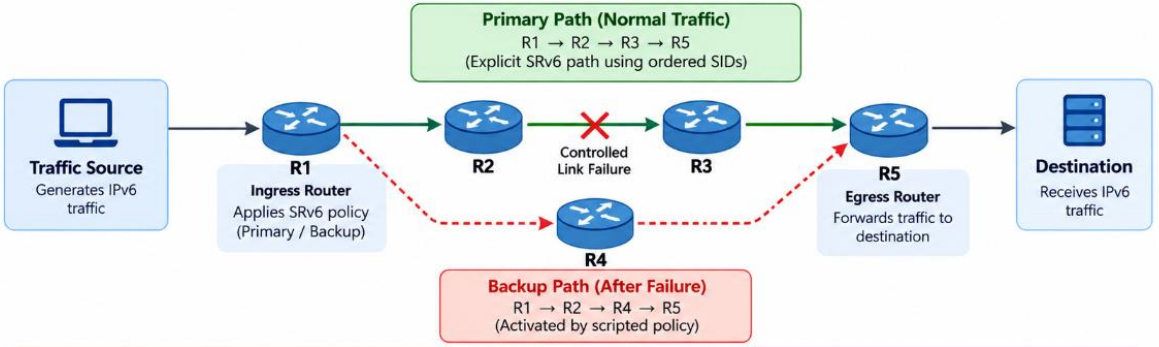
The proposed experiment improves path control, failure recovery, verification, and performance measurement compared with ordinary IPv6 routing.

These enhancements enable a structured comparison between ordinary IPv6 routing and SRv6 path steering under identical network conditions. Repeated failure trials will help obtain reliable measurements of recovery behaviour and network performance.

Aspect	Before	Proposed Experiment
Path Control	Ordinary IPv6 routing selects the path hop-by-hop.	Ingress applies an explicit SRv6 path policy using ordered SIDs.
Failure Response	Observe how the ordinary IPv6 baseline behaves under the same failure.	Detect the controlled failure and invoke the scripted backup policy.
Verification	Packet captures establish baseline traffic behaviour.	Packet captures verify SID traversal and the switched path.
Measurement	Measure packet loss and interruption under the same topology.	Additionally measure failure detection delay, repair delay, and SRv6 header overhead.

Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Failure Controller

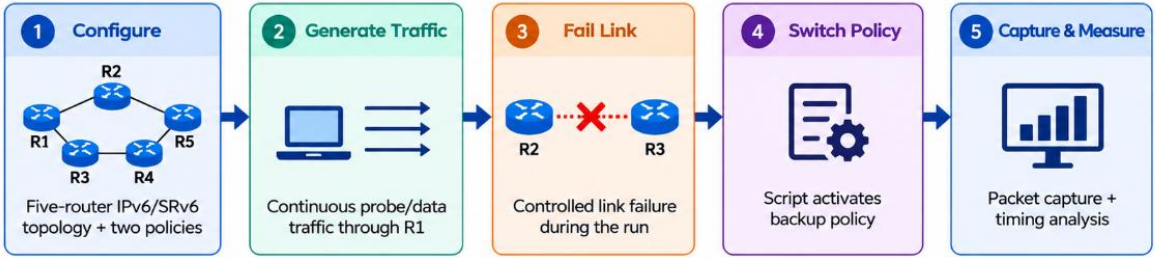
- Injects a controlled link failure (e.g., R2–R3).
- Triggers the scripted backup-policy switch on R1.

Measurement & Packet Capture

- Verifies SID traversal using packet captures.
- Measures packet loss, interruption time, failure detection delay, repair delay, and SRv6 header overhead.
- Compares SRv6 with ordinary IPv6 routing.

Data Flow Diagram

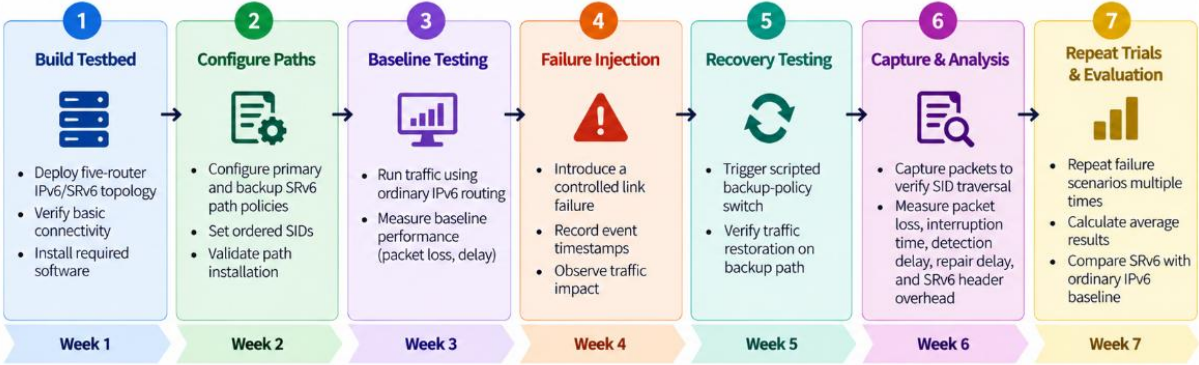
Overview of the architecture and flow of the SRv6 failure-recovery experiment.



- **Packet evidence:** Verify expected SID traversal in captured IPv6/SRv6 packets.
- **Timing evidence:** Timestamp failure, detection, policy switch and restored traffic.
- **Performance evidence:** Calculate packet loss and interruption time across repeated trials.
- **Overhead evidence:** Compare packet/header characteristics with ordinary IPv6 baseline.

Deployment Timeline

Project milestones and methodology for implementing and evaluating the SRv6 solution.



Methodology: Configure and validate testbed → Run baseline measurements → Apply controlled failure → Switch to backup policy → Capture and analyse results → Repeat trials and compare performance

Bibliography I

1. RFC 8754 — IPv6 Segment Routing Header (SRH)
Internet Engineering Task Force (IETF). *IPv6 Segment Routing Header (SRH)*. RFC 8754, March 2020. https://www.rfc-editor.org/rfc/rfc8754

2. AIORI-3 PS-020 Specification
AIORI-3 Hackathon. *Local SRv6 Path Steering and Measured Failure Recovery*. Problem Statement PS-020, Category: 6G & Future Networks.

Thank You